



Scan for the package,
the do-file, and this poster

Coding Open-Ended Survey Text Without Leaving Stata: the catllm Package for LLM-Assisted Content Coding

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Abstract

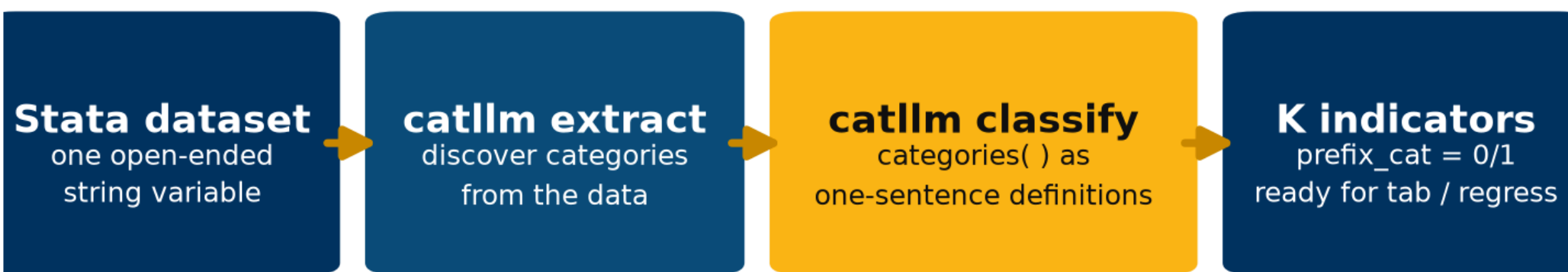
Open-ended survey items capture what closed-ended items cannot, and are routinely dropped from analysis because hand-coding them does not scale. `catllm` is a Stata command that codes free-text responses with large language models and returns ordinary Stata indicator variables. It wraps a Python backend (`cat-stack`) behind five verbs: `extract`/`explore` find categories, `classify` applies them, and `summarize`/`setup` close the loop, across OpenAI, Anthropic, Google, or Ollama. Tested on multi-label, multi-class items, proprietary models agree with human coders on **97%** of straightforward items and **88–91%** of complex interpretive ones; a multi-model ensemble reaches **98%** agreement with human consensus.

The Problem

- Researchers already use LLMs to code open-ended text, but little guidance exists for keeping classifications aligned with intended meaning without skewing conclusions.
- Model choice is often arbitrary. Researchers overpay for models that excel at general tasks, which rarely predicts coding accuracy.
- Coding several open-ended items at once means devising a bespoke strategy each time. That risks bias and error, not a documented method.
- Doing this from Stata isn't supported today: researchers export to CSV, script in Python, re-merge, and lose the audit trail that makes coding reproducible.
- Aim:** put the whole loop inside Stata, and tackle over-classification from several evidence-based angles: category definitions, model choice, and cross-provider ensemble consensus.

The Pipeline

Text goes in as a variable — categories come back as variables



Categories come from `extract` or your own list; `classify` returns multi-label indicator variables either way.

`generate()` is a prefix, not a variable name: a five-category scheme returns five 0/1 byte variables. `extract`'s specificity, iteration count, and category limit are tunable, so discovery matches the research question instead of a fixed default.

One Command

```
local cats ///
"Housing cost: Moved over rent, a mortgage, or affordability." ///
"Employment: Moved for a job, a transfer, or school." ///
"Family: Moved for caregiving or to be near relatives." ///
"Other: Does not fit any of the above categories."

catllm classify response, ///
categories(`cats') ///
apikey($OPENAI_API_KEY) ///
model("gpt-4o-mini") ///
generate(reason)

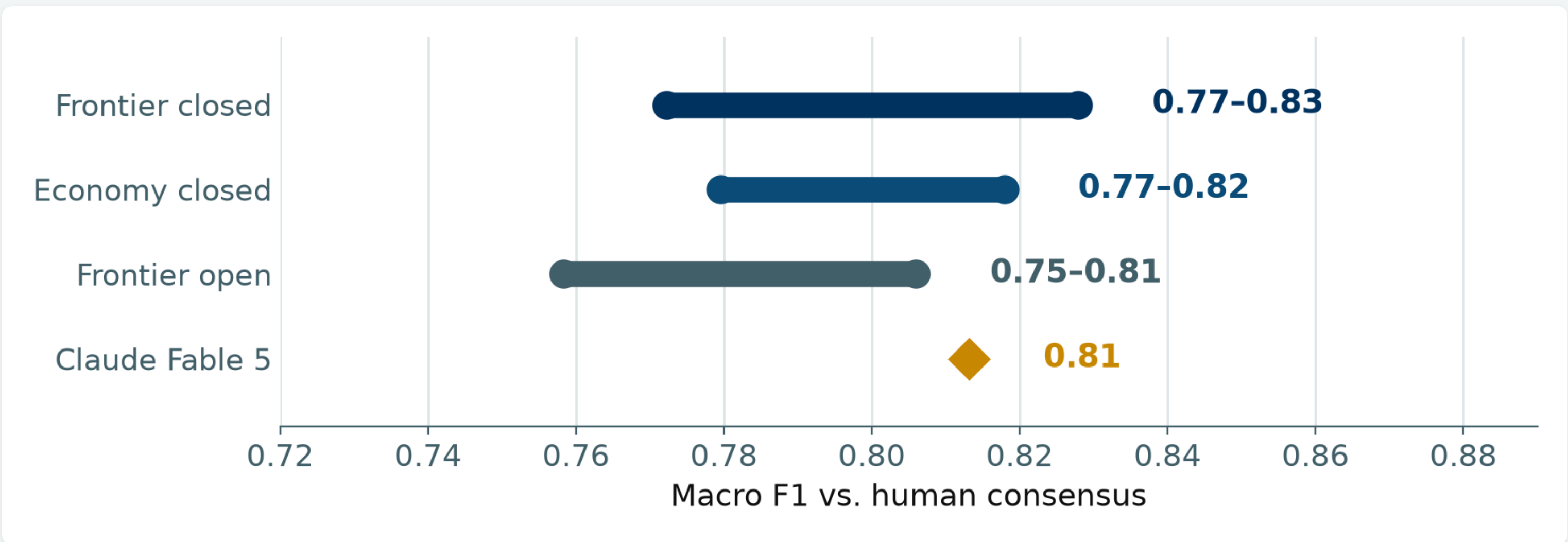
tab1 reason_*
regress reason_Housing_cost age female
```

Category definitions, not one-word labels: writing each category as a sentence classifies several percentage points better on every model tested. An explicit `other` category is required, not optional — without it the model must guess on responses that genuinely fit nothing.

References

Soria (2026). *CatLLM: A Python package for generating, assigning, and scoring open-ended survey data and images*. | Soria (2026). *High agreement, different stories*. | Soria (2026). *Model diversity over model size*. | Bojic et al. (2025). *Comparing LLM and human annotators*. | Wei et al. (2023). *Chain-of-thought prompting*. | Dhuliawala et al. (2023). *Chain-of-verification*. | Zheng et al. (2024). *Step-back prompting*.

Accuracy Against Human Coders

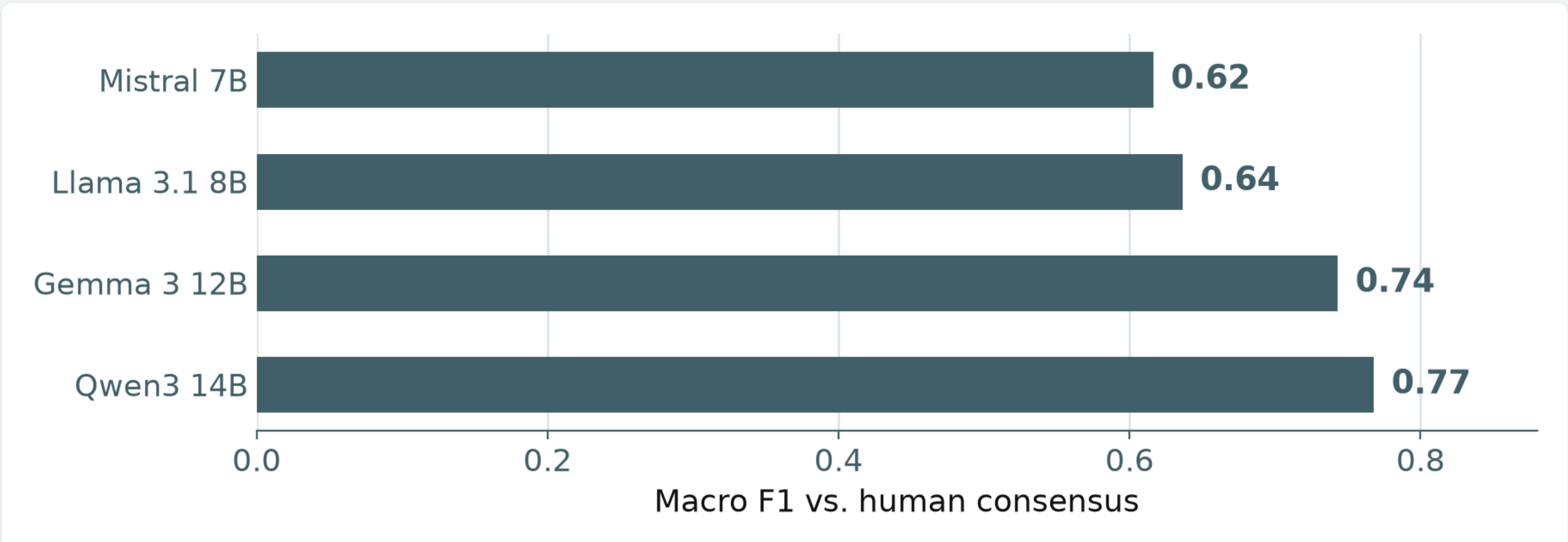


UC Berkeley Social Networks Study validation: macro F1 against human consensus, averaged across three survey questions so no single question dominates. Frontier-closed, economy-closed, and frontier-open tiers overlap closely; Claude Fable 5, shown individually, lands in line with the top cloud tiers. Ranges span the models within each tier.

What Moves Accuracy

- Cloud tier spread is narrow.** Frontier-closed, economy-closed, and frontier-open models span just 0.07 of macro F1 (0.75–0.83); Claude Fable 5 lands at 0.81, in line with the top tiers.
- Response brevity helps.** Responses under 50 characters are classified 7–11 points more accurately than long ones.
- Demographic variation is minimal.** Accuracy is stable across population segments — an essential precondition for using coded output in stratified analysis.
- Structured output is not the bottleneck.** Across 8 models × 3,208 responses (25,664 classifications), 100% of successful calls returned valid structured output; the 0–18 failures per model were transient server errors, not parsing failures.

Local Model Spread



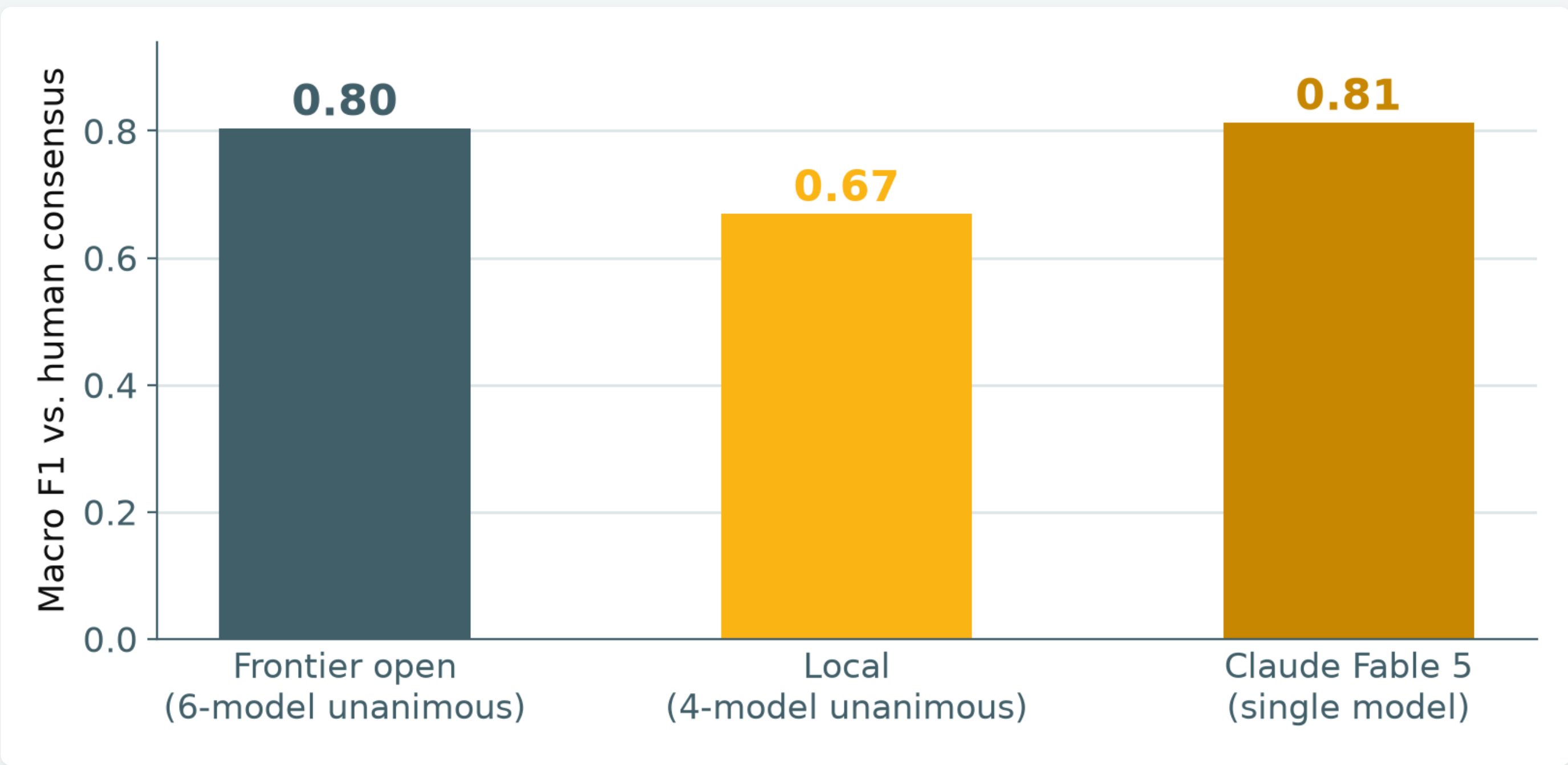
Same macro F1 against human consensus as the cloud tiers, for the four laptop-scale Ollama models. The spread here (0.62–0.77) is more than double the cloud tiers' (0.75–0.83) — picking well matters far more once you're local.

Choosing a Route

Route	Cost	Speed	Data leaves machine
Frontier proprietary	High	Slow	Yes
Small proprietary	Low	Fast	Yes
Multi-vendor ensemble	Sum of parts	Slowest member	Yes
Local open-weight (Ollama)	\$0	Hardware-bound	No

For restricted-use data under an IRB or data-use agreement, the local route is often the only admissible one — and it is the route the package defaults to protecting, auto-starting the local server and enabling JSON recovery for smaller models.

Unanimous Ensembles



consensus ("unanimous") requires every model in the ensemble to agree before a category is assigned. Combining all 6 frontier-open models this way nearly matches Claude Fable 5's single-model score; combining all 4 local models falls well short — unanimity can't make up for weaker members.

98%

ensemble vs.
human consensus

25,664

benchmark
classifications

\$0

fully local
route

Key Findings

. 0.75–0.83

Macro F1 against human consensus across cloud tiers; vendor and tier matter far less than the local-vs-cloud choice covered in *Choosing a Route*

. 98%

A three-model ensemble agrees with human consensus more often than any single model in it — disagreement between models is informative, not noise

. 0.62–0.77

Macro F1 across the four local Ollama models — more than double the cloud tiers' spread, so which local model you pick matters a lot

Reproduce This

Everything on this poster runs from one self-contained do-file — no data downloads, no file paths, guarded so it completes with a single API key or with none at all:

```
net install catllm, ///
from("https://raw.githubusercontent.com/chrissoria/cat-llm/main/stata-package/") ///
replace

catllm setup
do catllm_stata_conference.do
```

The file walks the full pipeline: classify, analyze the resulting indicators, compare three providers, run an ensemble with agreement scores, discover a coding scheme with `extract`, and repeat the whole thing locally at zero cost. Eight further worked examples ship with the package.

Limitations

- Concordance with human coders is not validity.** Where humans systematically misread a response, an agreeing model does too; agreement measures replication of a benchmark, not correctness.
- Validation is strongest on English-language survey text** of the kind studied here. Performance on other languages, registers, and domains needs separate checking on your own data.
- Models are moving targets.** A model version can be deprecated or silently updated, so the version string belongs in your methods section alongside the seed.
- Smaller local models are less tested.** The published benchmarks use high-end models; the low end warrants its own validation before use in production coding.